

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 8: Introduction to Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Ratios of standard angles

1. Given $\cos(\alpha + \beta) = 0$, find $\cos\left(\frac{\alpha + \beta}{2}\right)$. (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$ (c) 0 (d) $\sqrt{2}$ **CBSE 2024, 1 mark**

SOLUTION

$\cos(\alpha + \beta) = 0 = \cos 90^\circ$, so $\alpha + \beta = 90^\circ$.

$$\cos\left(\frac{\alpha + \beta}{2}\right) = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Answer: (a)

2. Find the value of $2\sqrt{2} \cos 45^\circ \sin 30^\circ + 2\sqrt{3} \cos 30^\circ$. **CBSE 2024, 2 marks**

SOLUTION

$$= 2\sqrt{2} \times \frac{1}{\sqrt{2}} \times \frac{1}{2} + 2\sqrt{3} \times \frac{\sqrt{3}}{2}$$

$$= 1 + 3$$

Answer: 4

3. For $A = 60^\circ$ and $B = 30^\circ$, verify that $\sin(A + B) = \sin A \cos B + \cos A \sin B$. **CBSE 2024, 2 marks**

SOLUTION

$$\text{LHS} = \sin(60^\circ + 30^\circ) = \sin 90^\circ = 1$$

$$\text{RHS} = \sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1$$

LHS = RHS. Verified.

2 Trigonometric identities

4. Given $\sec \theta - \tan \theta = m$, find $\sec \theta + \tan \theta$. (a) $1 - \frac{1}{m}$ (b) $m^2 - 1$ (c) $\frac{1}{m}$ (d) $-m$ **CBSE 2024, 1 mark**

SOLUTION

$$\sec^2 \theta - \tan^2 \theta = 1, \text{ so } (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1.$$

$$\sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta} = \frac{1}{m}$$

Answer: (c)

5. Prove the identity $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. **CBSE 2024, 3 marks**

SOLUTION

Write $s = \sin \theta$ and $c = \cos \theta$. Then $\frac{\tan \theta}{1 - \cot \theta} = \frac{s/c}{(s-c)/s} = \frac{s^2}{c(s-c)}$ and $\frac{\cot \theta}{1 - \tan \theta} = \frac{c/s}{(c-s)/c} = -\frac{c^2}{s(s-c)}$.

$$\begin{aligned} \text{LHS} &= \frac{s^3 - c^3}{sc(s-c)} = \frac{(s-c)(s^2 + sc + c^2)}{sc(s-c)} = \frac{s^2 + c^2 + sc}{sc} \\ &= \frac{1 + sc}{sc} = \frac{1}{sc} + 1 \\ &= 1 + \sec \theta \operatorname{cosec} \theta = \mathbf{RHS} \end{aligned}$$

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